

collate_ppg_dalia_dataset.m: MATLAB Data Collation Pipeline

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Toolbox / Script: collate_ppg_dalia_dataset.m & convert_subject_pickle_files_to_mat.py

Repository: <https://github.com/peterhcharlton/ppg-dalia-dataset>

Associated Projects: PPG-Beats, RRest, Wearable Physiological Monitoring Roadmaps.

1. Scientific Importance to this Capstone Project

In Section 2.2 of the Capstone Project report (*Đồ án Cuối kỳ AloT — Nguyễn Bách Tùng*), this script provides the definitive scientific benchmark confirming the distinction between target labels:

Data Field	Sampling Rate	Scientific Role & Target Definition	Alignment Strategy
raw.activity	4 Hz	Human Activity Recognition (HAR) Labels Annotates 8 discrete daily living activities.	Zero-Order Hold (ZOH) repeat interpolation (factor 8x) to match 32 Hz acceleration.
raw.label	0.5 Hz	Reference ECG Heart Rate (BPM) Computed every 2 seconds from chest ECG R-peaks.	Continuous physiological parameter; distinct from HAR classification targets.

2. Included Files in this Directory

- [03]_Charlton2026_collate_ppg_dalia_dataset.m: Full MATLAB script for data collation and ZOH alignment.
- [03]_Charlton2026_convert_subject_pickle_files_to_mat.py: Python conversion utility from raw pickle files to MAT structures.

Tài liệu đối chiếu phục vụ Đồ án Cuối kỳ AIoT — Sinh viên: Nguyễn Bách Tùng (MSSV: 23110166)

Trích dẫn: [3] P. H. Charlton, collate_ppg_dalia_dataset.m, University of Cambridge, 2026.